

Problem 3

Positional Encoding of NeRF:

```
freq_bands = torch.tensor(
    [2**i * torch.pi for i in range(1, self.num_octaves+1)],
    device=samples.device
)

...

embedded_samples[..., i + self.num_octaves :: 2 * self.num_octaves] = torch.cos(samples
```

NeRF one-line change:

```
radiance = self.alpha_composite(alphas, colors)
```

Short Answer

Our implementation uses uniform sampling, where the space between the near and far planes is divided into equal intervals; the NeRF paper uses stratified sampling (randomly sampling within intervals) and hierarchical sampling (refining sample locations based on density estimates). These methods in the original NeRF paper help improve rendering quality by focusing more samples in important regions, capturing fine details more effectively.



